

Heifer Completion Calculator - version 2.1

Installation instructions, updated 6/20/2026

<https://albertdevries.com/tools.html#heifercompletion>

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Introduction

Heifer completion refers to the fraction of born alive heifer calves that have a first calving themselves (approximately 2 years after birth). The remainder is sold or has died. This fraction is important for planning how many dairy heifers will be available to enter the milking herd. The Heifer Completion Calculator is an application that used DairyComp data and produces heifer completion results. The application is developed in Powershell, a shell program native in Windows, so that no new software needs to be installed on Windows computers. Input data are generated through 2 events and 1 list commands in DairyComp and saved automatically into CSV files. The application calculates completion fractions for 5 stages of heifer raising: Hutches (1 to 90 days old), Transition (91 to 180), Pre-breeding (181 to 370), Breeding (371 to 600), and Springer (601 to 800). The enrollment period is set to 180 days for each stage. Rather than calculating completion results for one cohort of calves from birth to calving (which would include only heifer calves born over 2 years ago), completion results are calculated independently for each stage using data from heifers that recently had the opportunity to complete their stage. The results from the 5 stages are strung together to provide a better estimate of current heifer completion fraction. Users can vary the ages in each stage, and the enrollment period. Results are shown both numerically and graphically and can be saved in Excel. The following steps show how to install and run the application. The heifer data are from the University of Florida.

Step 1:

Run the following 3 commands in DairyComp command line. Adjust your FOR statement filter to make sure only dairy heifers remain in the events and list (in the example below, dairy heifers have ID>10000). These 3 commands produce the HEIFER1.csv, HEIFER2.csv, and HEIFER3.csv files in the main DairyComp directory (for example, C:\DC305). Using Alter\3, you can put these commands into *command abbreviations*, for example *HEIFER1*, *HEIFER2* and *HEIFER3*:

HEIFER1:

```
SETUP\5RHEIFER1.CSV ! EVENTS ID LACT BDAT TODAY DIED SOLD FOR LACT=0  
ID>10000\2S1460 ! SETUP\4D
```

HEIFER2:

```
SETUP\5RHEIFER2.CSV ! EVENTS ID LACT BDAT TODAY FRESH FOR LACT=1\2S1460 !  
SETUP\4D
```

HEIFER3:

```
SETUP\5RHEIFER3.CSV ! LIST ID LACT BDAT TODAY RPRO DIM TODAY REM RC TECH FOR  
LACT=0 ID>10000 RC<>7\2S1460 ! SETUP\4D
```

Step 2:

Copy the two program files into the directory where the three files created in step 1 (HEIFER1.csv, HEIFER2.csv, HEIFER3.csv) are (or will be) located. The three csv files and the two program files must be in the same directory.

HEIFERCOMPLETION-BATCH.bat (calls HEIFERCOMPLETION-APP.exe)

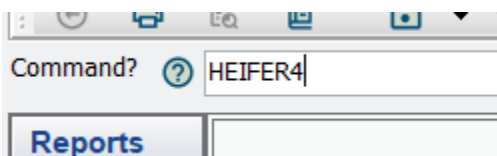
HEIFERCOMPLETION-APP.exe (compiled PowerShell script: calculations and results)

Name	File extension	Status	Date modified	Type	Size
 HEIFERCOMPLETION-APP	.exe		6/19/2026 10:50 PM	Application	134 KB
 HEIFERCOMPLETION-BATCH	.bat		5/5/2026 11:23 PM	Windows Batch File	1 KB

Using Alter\3, you can make a *command abbreviation* and run the BATCH file directly from the DairyComp command line, for example *HEIFER4*:

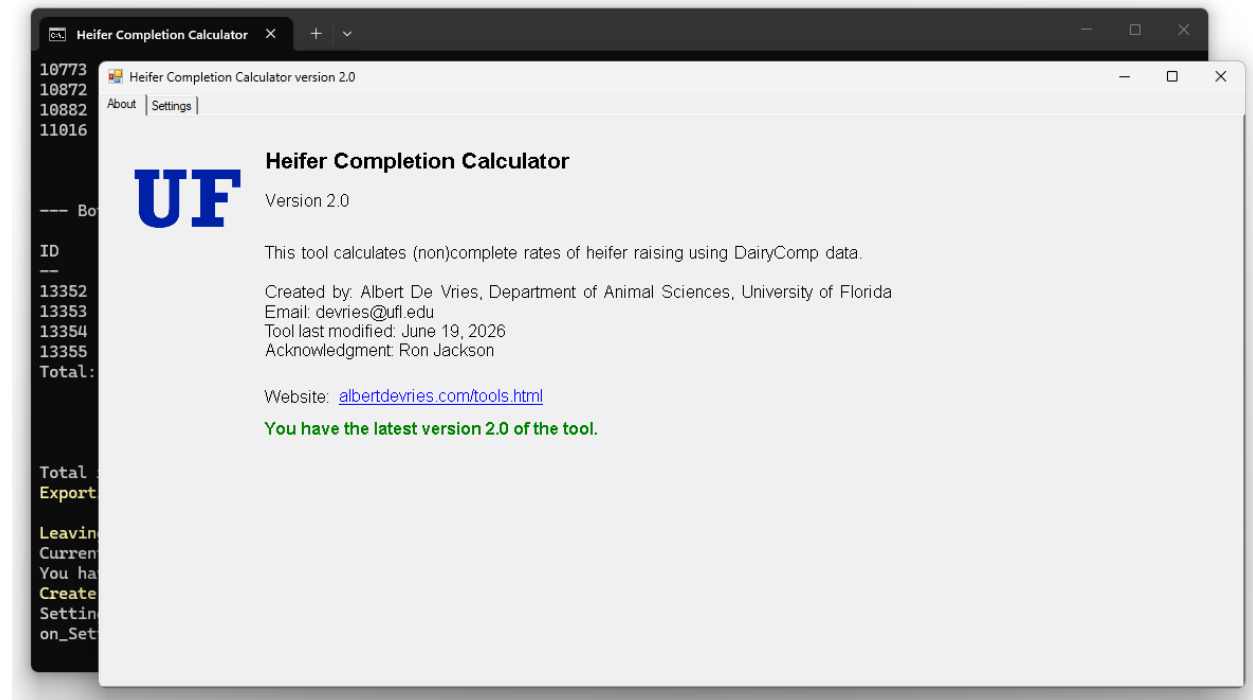
HEIFER4:

```
EXEC\BHEIFERCOMPLETION-BATCH
```



Step 3:

Run the BATCH file (for example by double clicking). This should call the HEIFERCOMPLETION-APP.exe which reads the three csv files and opens the user-interface, usually on the Settings tab and sometimes on the About tab. The About tab is activated when a new version is available at <https://albertdevries.com/tools.html>



Step 4:

Change the yellow cells in the Settings tab of the form or leave the default values. Click the red **Calculate Results** button. You can save the values in the yellow cells by clicking the green **Save Settings** button.

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About Settings

Enter Your Own Settings

Stage	End Age (days)	Enrollment (days)
Hutches	90 1 to 90 days	180 days
Transition	180 91 to 180 days	180 days
Prebreeding	365 181 to 365 days	180 days
Breeding	600 366 to 600 days	180 days
Springer	800 601 to 800 days	180 days

Calculate Results **Save Settings**

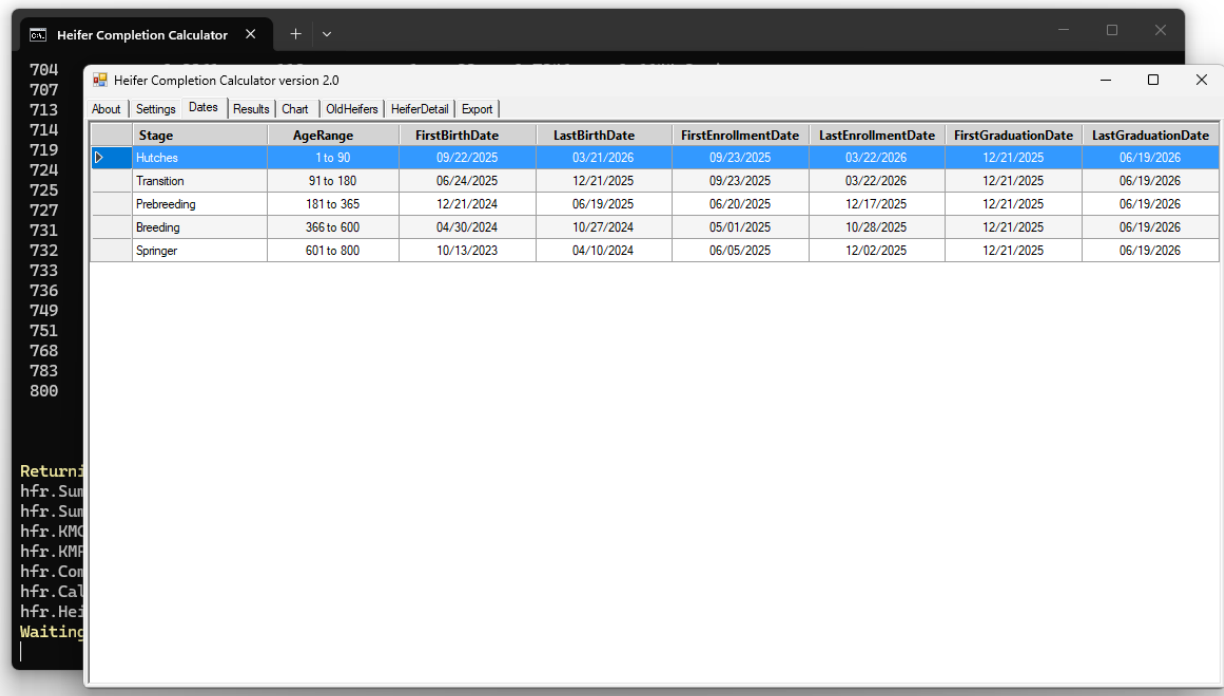
Click "Calculate Results" to calculate and view Dates, Results, Chart, OldHeifers, HeiferDetail and Funnel.

Heifer files last modified dates:
heifer1.csv: Friday, June 19, 2026 8:23 PM
heifer2.csv: Friday, June 19, 2026 8:24 PM
heifer3.csv: Friday, June 19, 2026 8:25 PM

Step 5:

See output in the Dates tab, Results tab, Chart tab, OldHeifers tab and HeiferDetail tab. The Export tab saves all input and output in an Excel file. Go back to the Settings tab and change yellow cells if needed. Click X in the top right corner of the form to exit the application.

The **Dates tab** shows the age range, first and last birthdates in the state, the first and last enrollment in the stage, and the first and last graduation date. Heifers that survive the stage are said to graduate from the stage.



The screenshot shows the 'Heifer Completion Calculator' application window. The 'Dates' tab is selected, displaying a table with the following data:

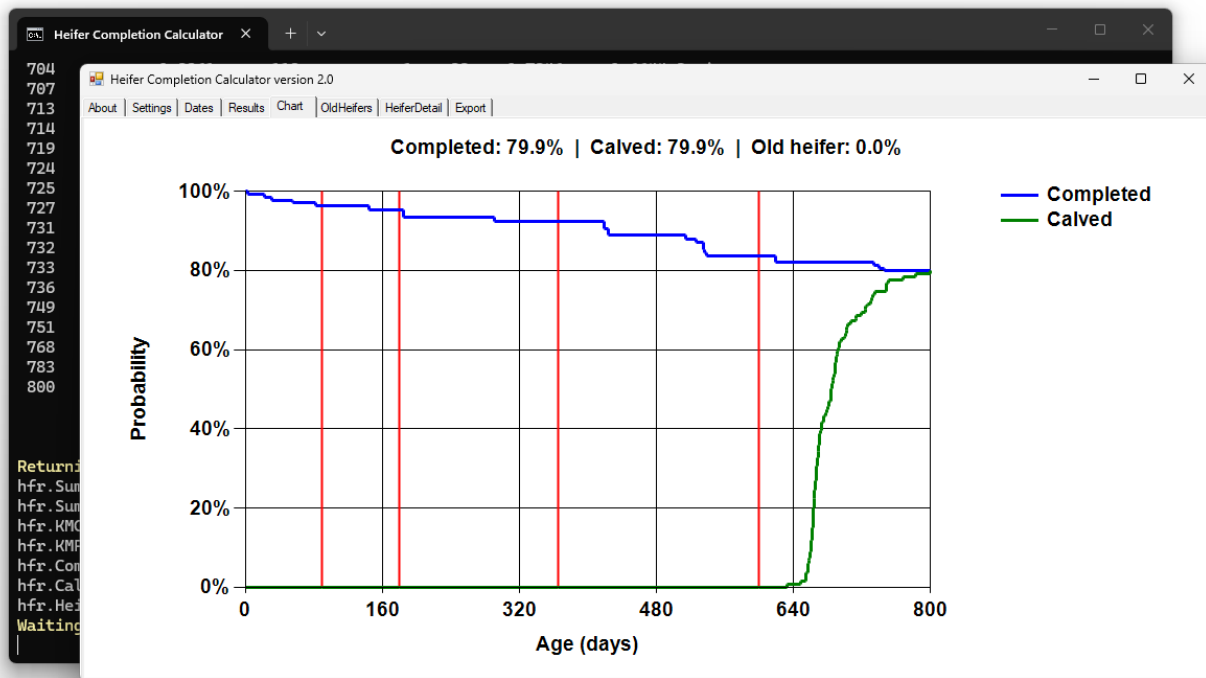
Stage	AgeRange	FirstBirthDate	LastBirthDate	FirstEnrollmentDate	LastEnrollmentDate	FirstGraduationDate	LastGraduationDate
Hutches	1 to 90	09/22/2025	03/21/2026	09/23/2025	03/22/2026	12/21/2025	06/19/2026
Transition	91 to 180	06/24/2025	12/21/2025	09/23/2025	03/22/2026	12/21/2025	06/19/2026
Prebreeding	181 to 365	12/21/2024	06/19/2025	06/20/2025	12/17/2025	12/21/2025	06/19/2026
Breeding	366 to 600	04/30/2024	10/27/2024	05/01/2025	10/28/2025	12/21/2025	06/19/2026
Springer	601 to 800	10/13/2023	04/10/2024	06/05/2025	12/02/2025	12/21/2025	06/19/2026

On the left side of the application, a list of heifer names is visible, including: 704, 707, 713, 714, 719, 724, 725, 727, 731, 732, 733, 736, 749, 751, 768, 783, 800, Return, hfr. Sum, hfr. Sum, hfr. KMQ, hfr. KMF, hfr. Com, hfr. Cal, hfr. He3, and Waiting.

The **Results tab** shows the counts and percentages in the stages. The Completed% column is the multiplication of the Graduated% stages. The Completed% in the Springer stage is the estimate of the heifer completion fraction from birth to calving. The StillHeifer% include the heifers that have not calved (freshened) at the end of the Springer stage. They are usually old heifers, perhaps heifers with incomplete records.

Stage	Enrolled	Died	Sold	Fresh	Graduated	Died%	Sold%	Graduated%	Completed%	Calved%	StillHeifer%
Hutches	136	5	0	0	131	3.7%	0.0%	96.3%	96.3%	0.0%	96.3%
Transition	97	1	0	0	96	1.0%	0.0%	99.0%	95.3%	0.0%	95.3%
Prebreeding	98	0	3	0	95	0.0%	3.1%	96.9%	92.4%	0.0%	92.4%
Breeding	105	0	10	0	95	0.0%	9.5%	90.5%	83.6%	0.0%	83.6%
Springer	112	2	3	107	0	1.8%	2.7%	0.0%	79.9%	79.9%	0.0%

The **Chart tab** shows the composite survival curve (in blue) and the composite calved curve (in green). The difference between the Completed and Calved percentages is the percentage of Old Heifers.



The **OldHeifers** tab is not shown because there were no old heifers in the data. The **HeiferDetail** tab is a list of the heifers and their status included in each stage.

ID	Stage	Outcome	BirthDate	EnrollmentDate	EventDate	Source
13161	Hutches	Died	09/24/2025	09/25/2025	11/18/2025	heifer1.csv
13178	Hutches	Died	10/11/2025	10/12/2025	01/01/2026	heifer1.csv
13206	Hutches	Died	11/08/2025	11/09/2025	12/01/2025	heifer1.csv
13236	Hutches	Died	01/06/2026	01/07/2026	02/06/2026	heifer1.csv
13250	Hutches	Died	01/19/2026	01/20/2026	01/22/2026	heifer1.csv
13162	Hutches	Graduated	09/24/2025	09/25/2025	06/19/2026	heifer3.csv
13163	Hutches	Graduated	09/26/2025	09/27/2025	06/19/2026	heifer3.csv
13164	Hutches	Graduated	09/26/2025	09/27/2025	06/19/2026	heifer3.csv
13165	Hutches	Graduated	09/27/2025	09/28/2025	06/19/2026	heifer3.csv
13166	Hutches	Graduated	09/27/2025	09/28/2025	06/19/2026	heifer3.csv
13167	Hutches	Graduated	09/27/2025	09/28/2025	06/19/2026	heifer3.csv
13168	Hutches	Graduated	10/01/2025	10/02/2025	06/19/2026	heifer3.csv
13169	Hutches	Graduated	10/04/2025	10/05/2025	06/19/2026	heifer3.csv
13170	Hutches	Graduated	10/04/2025	10/05/2025	06/19/2026	heifer3.csv
13171	Hutches	Graduated	10/04/2025	10/05/2025	06/19/2026	heifer3.csv
13172	Hutches	Graduated	10/05/2025	10/06/2025	06/19/2026	heifer3.csv
13173	Hutches	Graduated	10/05/2025	10/06/2025	06/19/2026	heifer3.csv
13174	Hutches	Graduated	10/06/2025	10/07/2025	06/19/2026	heifer3.csv
13175	Hutches	Graduated	10/07/2025	10/08/2025	06/19/2026	heifer3.csv
13176	Hutches	Graduated	10/08/2025	10/09/2025	06/19/2026	heifer3.csv
13177	Hutches	Graduated	10/11/2025	10/12/2025	06/19/2026	heifer3.csv

The **Export tab** saves the output in an Excel spreadsheet.

